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**Dense Drawdown Supervision for Equity Crisis  
Early Warning: A Multi-Task Temporal Fusion  
Transformer and an Honest Test of Regime Awareness**

*MSc Dissertation*

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## **Declaration**

I declare that this dissertation is my own work. All experiments, code, data processing, and results reported here were produced by me, except where explicitly attributed to others through citation. The data are daily Bloomberg market series; the modelling code, experiment scripts, and figure-generation code are my own and are available in the project repository (see Appendix A).

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## Abstract

Equity crisis early warning systems (EWS) are usually built on a single market, where only a handful of crisis episodes occur per decade. In that low-data regime gradient-boosted trees are widely reported to match or beat neural sequence models, and it is tempting to read any deep-learning gain as a triumph of architecture. We test that reading. Taking the Temporal Fusion Transformer (TFT) of Lim et al. (2021) as a backbone, we ask two questions under a strict leak-free protocol: does *regime awareness*, a component that tries to infer for itself whether the market is calm, stressed, or in crisis and re-weight the model accordingly, help? And if not, what does? Across six crisis-anchored walk-forward folds with a 63-day embargo and thirty seeds, tested at the level of the crisis episode, regime awareness does nothing: the regime-aware TFT is statistically indistinguishable from a vanilla one. What helps is duller, and lives in the supervision signal rather than the architecture. Alongside the rare yes/no crisis label we train the model to predict how far the market will fall over the coming quarter, a target defined on every day rather than only at the sparse onsets, and add a *focal loss* that concentrates on the hard cases. We call the pair the “AF” recipe. Its most consistent payoff is calibration: AF lowers the Brier score in all six folds. Discrimination (PR-AUC, read against a 0.34 no-skill floor) rises by about 0.03 on average, concentrated in the severe COVID and China-2015 crises, though with only six episodes it falls short of significance. Holding the objective fixed and ablating the two heads separates the ingredients, and the result is sharper than the recipe-level comparison: the auxiliary drawdown head improves PR-AUC in *every* one of the six folds (macro +0.0342; episode-level Wilcoxon  $p = 0.0312$ , bootstrap CI [+0.0133, +0.0595]), while the focal loss contributes nothing on its own. The dense drawdown head, not the loss, is the contribution. On a nine-market panel ( $\approx 110$  onsets), with capacity matched to the larger data, the tuned recipe (“AFv2”) edges XGBoost (PR-AUC 0.3846 vs. 0.3728, best Brier), and a five-fold panel walk-forward keeps that ordering; the same configuration transfers back to the single market unchanged. The story is as much about method as model: a careful, reproducible, leak-free evaluation, with tuned tree baselines, calibration and usefulness analysis, and walk-forward checks, that separates a real improvement (dense drawdown supervision) both from a plausible one the evidence does not support (regime modules) and from an inert one it would have been easy to credit (the focal loss).

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## Abbreviations and Notation

### Abbreviations

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|                 |                                                        |
|-----------------|--------------------------------------------------------|
| A100            | NVIDIA A100, the GPU model used for all training runs  |
| AF              | auxiliary forward-drawdown head + focal loss           |
| AFv2            | AF with panel-matched capacity (see Sec. 5.7)          |
| BCS             | British Computer Society                               |
| dd10            | 10% drawdown crisis threshold                          |
| DXY             | US dollar index                                        |
| ECE             | expected calibration error                             |
| EWS             | early warning system                                   |
| GFC             | global financial crisis (2008)                         |
| GRN             | gated residual network                                 |
| HMM             | hidden Markov model                                    |
| HPC             | high-performance computing                             |
| IET             | Institution of Engineering and Technology              |
| MCX, MXEF, MXWO | FTSE 250, MSCI EM, MSCI World                          |
| MLP             | multilayer perceptron                                  |
| MOVE            | bond-volatility index                                  |
| PR-AUC          | area under the precision–recall curve                  |
| RA-TFT          | regime-aware TFT                                       |
| ROC-AUC         | area under the receiver-operating-characteristic curve |
| Slurm           | the cluster job scheduler used on KCL CREATE           |
| SPX, NDX, RTY   | S&P 500, Nasdaq 100, Russell 2000                      |
| SX5E, SXXP, UKX | EuroStoxx 50, Stoxx 600, FTSE 100                      |
| TED             | TED spread (interbank minus Treasury rate)             |
| TFT             | Temporal Fusion Transformer (Lim et al., 2021)         |
| VIX             | equity-volatility index                                |
| VSN             | variable selection network                             |

### Notation

---

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| $X_{m,t}$   | closing price of market $m$ on trading day $t$                          |
| $P_{m,t}$   | trailing 252-day rolling peak of $X_{m,t}$                              |
| $D_{m,t}$   | running drawdown $X_{m,t}/P_{m,t} - 1 \leq 0$                           |
| $\tau$      | crisis drawdown threshold ( $\tau = 0.10$ for dd10)                     |
| $y_{m,t}$   | binary EWS onset label over $[t+1, t+63]$                               |
| $d_{m,t,h}$ | forward running drawdown $D_{m,t+h}$ , $h = 1, \dots, 63$               |
| $p$         | predicted onset probability; $\gamma$ focal focusing; $w_+$ pos. weight |
| $\pi_t$     | regime posterior over $K$ states at step $t$                            |
| $\pi$       | label prevalence $P/(P+N)$ (PR-AUC baseline)                            |

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# 1 Introduction

## 1.1 Domain and problem

Equity bear markets impose large, asymmetric costs: capital is committed near peak valuations, and credit tightens precisely when firms and households can least absorb it. A reliable *early warning system* (EWS) for crisis onset is therefore of direct interest to risk managers, allocators, and macroprudential authorities (Kaminsky and Reinhart, 1999; Reinhart and Rogoff, 2009). We treat crisis prediction as a binary classification problem: given the market state up to day  $t$ , predict whether a *drawdown* crisis, a sustained fall from the market’s recent price peak, will *begin* within the next quarter, i.e. in the window  $[t + 1, t + 63]$  trading days.

## 1.2 The single-market trap and a tempting hypothesis

A persistent finding in the EWS literature is that gradient-boosted trees match or beat neural sequence models (Bluwstein et al., 2023; Holopainen and Sarlin, 2017; Beutel et al., 2019). A natural reading is that crisis prediction is a fundamentally low-data problem: a single index produces only  $\sim 4\text{--}6$  distinct crises in a quarter-century, far too few for an attention layer with  $O(d^2)$  parameters to learn against. This invites two responses. The first, and the one we initially pursued, is to add a crisis-specific inductive bias to the network: a *regime-aware* TFT (RA-TFT) with a differentiable hidden-Markov regime detector and a regime-conditioned attention bias, on the intuition that a model which “knows” it is in a stressed regime should attend to history differently. The second is to attack the data scarcity directly, by pooling many markets into a panel.

## 1.3 Aims and objectives

The **aim** of this project is to establish whether regime-awareness improves transformer-based early warning of equity drawdown crises once the evaluation is made genuinely leak-free and, if it does not, to identify what does. The supporting **objectives** are:

1. Construct a leak-free evaluation protocol for daily equity crisis EWS: crisis-anchored walk-forward folds, a temporal embargo matched to the label horizon, strictly post-split windowing, and inference at the level of the crisis episode rather than the random seed.
2. Establish a fair backbone comparison on those splits between the TFT, a recurrent baseline, and hyperparameter-tuned gradient-boosted trees, the models the literature reports as hardest to beat.
3. Implement the regime-aware TFT in full (a differentiable neural-HMM regime module, a regime-conditioned attention bias, and an auxiliary regime loss), and ablate it component by component rather than as a single block.
4. Attack the rare-event scarcity through the *learning problem* instead of the architecture, by adding dense auxiliary supervision of the forward drawdown and a focal loss over the sparse onset label.
5. Test whether the resulting recipe transfers to a multi-market panel, where crisis episodes are an order of magnitude more numerous.
6. Evaluate probability *calibration* and decision usefulness alongside discrimination, since an early-warning probability is only actionable if it can be believed.

## 1.4 What we actually found, and how the project was re-framed

Two results stand out, one negative and one positive. The negative one came first. Under the leak-free protocol the regime machinery does not move the needle: a fully regime-aware TFT scores no better than a plain one (Section 5.2), on the single market and again on the panel. The positive result is simpler, and not architectural at all. Alongside the rare yes/no crisis label, we also train the model to predict how far the market falls over the coming quarter, a target defined on every day rather than only at the sparse onsets, and we pair it with a focal loss that keeps the gradient on the hard cases. We call this the *AF* recipe. On the single market it beats the vanilla TFT, most visibly in the calibration of its probabilities; scaled up to a nine-market panel with capacity matched to the data (the *AFv2* configuration) it edges past XGBoost, the baseline the literature keeps warning about.

The project is therefore built around the recipe, not the regime module. We still report the regime work in full. It was a reasonable idea, tested carefully, that did not work out, and the leak-free machinery that buried it is what makes the positive result believable.

## 1.5 Contributions

1. **A leak-free evaluation protocol** for daily equity crisis EWS: crisis-anchored walk-forward folds, a 63-day temporal embargo matched to the label horizon (López de Prado, 2018), strictly post-split windowing, train-only feature scaling, PR-AUC reported *relative to the prevalence baseline*, calibration analysis (Brier, ECE, reliability diagrams, Platt scaling), the Sarlin (2013) cost-sensitive usefulness measure, episode-level significance testing with block-bootstrap confidence intervals, and a tuned multi-model tree-baseline comparison. We show that under this protocol an earlier, random-holdout result reverses, a cautionary, reproducible demonstration of how much the evaluation protocol shapes the conclusion.
2. **A powered, factorial test of regime awareness.** A differentiable neural-HMM regime module and regime-conditioned attention bias are implemented and ablated component by component. They are *inert*: a fully regime-aware TFT is statistically indistinguishable from a vanilla one, and adding seeds does not rescue it (Section 5.2).
3. **Dense drawdown supervision, isolated from the loss.** A multi-task auxiliary forward-drawdown regression head plus focal loss (the “AF” recipe) *consistently improves* on the vanilla TFT on the single-market task: it lowers the Brier score in all six crisis folds and raises macro PR-AUC (0.4869 vs. 0.4592), with the discrimination gain concentrated in severe crises and, at the episode level, short of conventional significance. A three-way head ablation with the objective held fixed then shows that the auxiliary head carries the effect on its own, improving PR-AUC in all six folds (macro +0.0342; Wilcoxon  $p = 0.0312$ ; bootstrap CI excluding zero), whereas the focal loss is inert. We therefore attribute the gain to the dense target rather than to the reweighted loss (Section 5.4).
4. **A panel where the tuned recipe is competitive with gradient boosting.** On a nine-market Bloomberg panel, the capacity-matched AFv2 configuration is marginally ahead of XGBoost on PR-AUC and calibration (0.3846 vs. 0.3728 PR-AUC; best Brier), and the same configuration transfers back to the single-market task without loss, a single, universal recipe. We validate this against *tuned* tree baselines (XGBoost, Random Forest, histogram gradient boosting) and a five-fold panel walk-forward, both of which preserve the ordering.
5. **A reproducible pipeline** from raw Bloomberg series through calibrated probabilistic forecasts, with all results regenerated by Slurm array jobs on HPC.

## 1.6 Structure of this report

Section 2 reviews the crisis early-warning literature, the sequence models this work builds on, and the evaluation pitfalls specific to financial time series. Section 3 describes the Bloomberg panel and the features derived from it. Section 4 is the core of the approach: how crises are labelled and why the 10% threshold is defensible, how the crisis-anchored walk-forward folds and the embargo remove leakage, and how the model, the AF recipe and the regime branch are constructed. Section 5 reports the experiments: the backbone comparison, the regime ablation, the AF result, calibration, threshold sensitivity, and the panel. Section 6 interprets these and states the limitations, Section 7 discusses the legal, social, ethical and professional dimensions of the work, and Section 8 concludes with the lessons learned and the directions we would pursue next.

## 2 Background and Related Work

### 2.1 Crisis early warning systems

The modern EWS literature begins with the signals approach of Kaminsky and Reinhart (1999) and matured into a logistic-regression tradition (Frankel and Saravelos, 2012) and then a machine-learning tradition (Beutel et al., 2019; Holopainen and Sarlin, 2017; Chatzis et al., 2018). A robust empirical regularity, crystallised by Bluwstein et al. (2023) on a long credit-cycle panel, is that ensemble trees match or beat neural models. The implicit explanation is the low-data inductive-bias advantage of trees. Our results are consistent with this on single markets but show it is not a law: with enough episodes and capacity matched to the data, a properly trained TFT overtakes XGBoost.

A second strand concerns the *label*. Naive drawdown regression conflates recovery from a prior crash with the onset of a new one. The accepted remedy is binary *onset* labelling over a fixed forward horizon, excluding in-crisis observations from training (Dichtl et al., 2023; Holopainen and Sarlin, 2017; Bluwstein et al., 2023). The drawdown itself should be measured from a *trailing local peak*, not an all-time high, which is the convention in business-cycle and bull/bear dating (Pagan and Sossounov, 2003; Lunde and Timmermann, 2004). We adopt both.

### 2.2 Sequence models, TFT, and the data-to-parameter ratio

Recurrent and attention architectures excel where data are abundant (limit-order-book, intraday volatility, retail demand) but underperform on single-market crisis prediction, where the sample-to-parameter ratio is adverse. The Temporal Fusion Transformer (Lim et al., 2021) is purpose-built for *panel* forecasting: variable selection networks, gated residual connections, interpretable multi-head attention, and, crucially, a static covariate encoder that learns entity embeddings and conditions the rest of the network. That static pathway is exactly what lets a single model share temporal structure across markets while preserving market identity, which is why TFT is the right backbone once the problem is posed as a panel.

### 2.3 Class imbalance, multi-task learning, and the AF recipe

Crisis labels are rare and bursty. Two standard tools address this. *Focal loss* (Lin et al., 2017) down-weights easy negatives so the gradient concentrates on the hard, rare positives. *Multi-task learning* adds an auxiliary objective whose dense supervisory signal regularises the shared representation; here the auxiliary target is the continuous forward-drawdown magnitude, which is defined on *every* day rather than only at the rare onsets. The combination is the core of our

positive result. Deep models with auxiliary heads and modern losses are established in empirical asset pricing (Gu et al., 2020).

## 2.4 Regime switching

Regime-switching models are a workhorse of financial econometrics since Hamilton (1989), with heavy use in volatility and allocation (Ang and Bekaert, 2002). Their neural-HMM incarnation (Tran et al., 2016) keeps the forward algorithm differentiable so the regime layer can be trained end-to-end inside a larger network. The hypothesis that such a layer should help crisis EWS is reasonable a priori; our contribution is to test it rigorously and report that, in this setting, it does not.

## 2.5 Evaluation and leakage in financial time series

For rare-event detection, PR-AUC is the appropriate primary metric and is more informative than ROC-AUC (Saito and Rehmsmeier, 2015; Davis and Goadrich, 2006); it should be read against the prevalence baseline  $P/(P+N)$ , not 0.5. Probability *calibration* (Brier score, reliability) matters because EWS outputs feed decisions; Platt scaling is preferred to isotonic regression on small validation sets (Niculescu-Mizil and Caruana, 2005). Leakage control is the dominant threat: financial cross-validation requires *purging* and an *embargo* at least as long as the label horizon (López de Prado, 2018), and naive time-series CV is unreliable (Bergmeir and Benítez, 2012). Our protocol implements all of these (Section 4.2).

## 2.6 The gap

Single-market crisis EWS is data-starved and trees dominate, but this is a property of the *formulation*. TFT is the natural panel backbone yet is under-used for crisis EWS. Regime switching is well established but its value as a differentiable layer for crisis prediction is asserted more often than tested. This dissertation closes the last gap with a negative result and fills the first two with a positive, transferable recipe.

# 3 Data

## 3.1 Source and instruments

All series are daily Bloomberg data, 3 January 2000 to May 2026. The single-market study uses the S&P 500 (SPX). The panel study uses nine broad equity indices with full coverage over the period, drawn from one consistent source: S&P 500 (SPX), Nasdaq 100 (NDX), Russell 2000 (RTY), EuroStoxx 50 (SX5E), Stoxx 600 (SXXP), FTSE 100 (UKX), FTSE 250 (MCX), MSCI Emerging Markets (MXEF), and MSCI World (MXWO). The panel contains 61,996 market-days and approximately 110 distinct crisis-onset episodes, against the  $\sim 4$ –6 isolated on the S&P 500 alone, the statistical-power gain that the panel reformulation buys.

## 3.2 A note on static covariates

The dataset contains *no time-invariant static numeric variables*. (The “\_Static” Bloomberg files are a one-shot *pull type*, not static features; every series in the manifest is a daily time series.) Consequently, on the single-market task TFT’s static-covariate pathway is inherently dormant: there is exactly one constant market identifier and no static numerics. This is a property of single-asset EWS, not a defect. The pathway becomes meaningful only in the panel, where *market*

*identity* is a genuine static categorical with nine levels, consumed by TFT’s static encoder as a learned market embedding.

### 3.3 Features

Each input is a 252-day history window ending at  $t$ . Per-market features (23 historical inputs) are: log returns at 1/5/21 days; realised volatility at 5/21/63 days; 63-day drawdown from the rolling peak; 63-day vol-of-vol, return skewness, and Hurst exponent; 21-day Amihud illiquidity; 63-day volume  $z$ -score; and the macro-financial block VIX, US high-yield and investment-grade credit spreads and their difference, the TED spread, gold, the dollar index (DXY), the MOVE index, the implied–realised volatility spread, a gold/SPX ratio, and 5-day DXY return. Two *known-future* calendar features (sine/cosine of the annual cycle) form the decoder inputs. Non-stationary level series enter only through returns or are standardised by the train-only scaler. All rolling statistics are causal (backward windows only).

## 4 Methodology

### 4.1 Crisis labelling and the dd10 threshold

For market  $m$  and day  $t$  let  $P_{m,t} = \max_{t-251 \leq s \leq t} X_{m,s}$  be the trailing 252-day rolling peak of the closing price  $X$  (the highest close in the past trading year), and let the running drawdown be  $D_{m,t} = X_{m,t}/P_{m,t} - 1 \leq 0$  (how far below that peak the market currently sits). Market  $m$  is *in crisis* on day  $t$  if the drop is at least  $\tau$ , i.e.  $|D_{m,t}| \geq \tau$ . A crisis *onset* is the *first* day a new crisis begins, the first in-crisis day preceded by at least 20 calm days, a *hysteresis* buffer so that one long crash is not miscounted as many separate crises. (“dd10” below denotes the choice  $\tau = 10\%$ .) All windows in  $[t+1, t+63]$  are measured in *trading* days. The binary EWS target is

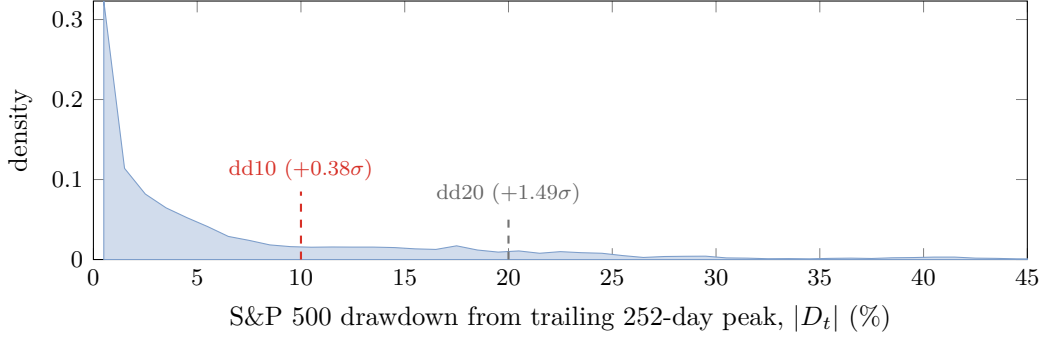
$$y_{m,t} = \mathbf{1}\{\exists s \in [t+1, t+63] \text{ such that } s \text{ is a crisis onset}\}. \quad (1)$$

In-crisis observations are excluded from *training* so the model learns to anticipate onsets rather than recognise an ongoing crash; they are retained at *test* time so evaluation reflects deployment, where the regime is unknown a priori. Days in the final 63 of the dataset (whose forward window is incomplete) receive no label and are dropped.

**Why  $\tau = 10\%$  (“dd10”).** The threshold is chosen from the *empirical* drawdown distribution rather than convention. Figure 1 shows the distribution of  $|D_t|$  on the S&P 500 (mean 6.61%, standard deviation 9.00%). Table 1 translates candidate thresholds into tail position and exceedance frequency. A 5% threshold sits *below* the mean drawdown ( $-0.18\sigma$ , exceeded on 36% of days): it is noise, not crisis. A 20% threshold is statistically clean ( $+1.49\sigma$ , 9.4% of days) but yields only  $\sim 5$  episodes in 26 years, too few for multi-seed walk-forward evaluation. We therefore use  $\tau = 10\%$  ( $+0.38\sigma$ , 23.6% of days) and *name the task honestly* as a market-stress / correction EWS rather than a deep-crisis EWS. The sensitivity of conclusions to  $\tau$  is reported in Section 5.6.

### 4.2 Crisis episodes and the walk-forward protocol

Figure 2 plots the S&P 500 with its crisis episodes shaded. Evaluation is *crisis-anchored walk-forward*: each fold trains on an expanding history, validates on a calm window, and tests on a window containing a target crisis. The six folds are GFC (2008), Eurozone (2011), China (2015), the Q4-2018 sell-off, COVID (2020), and the 2022 bear market.



**Figure 1:** Empirical distribution of S&P 500 drawdowns from the trailing 252-day peak (2000–2026). The chosen 10% threshold lies just above the mean drawdown; 20% sits well into the tail but is too rare to support walk-forward evaluation.

**Table 1:** Drawdown thresholds on the S&P 500: tail position ( $z$  on  $|D_t|$ , mean 6.61%, s.d. 9.00%), exceedance frequency, and approximate episode count. dd10 is the chosen operating point.

| Threshold $\tau$  | $z = (\tau - \mu)/\sigma$ | % of days $\geq \tau$ | approx. episodes |
|-------------------|---------------------------|-----------------------|------------------|
| 5%                | −0.18                     | 36.4%                 | many (noise)     |
| <b>10% (dd10)</b> | <b>+0.38</b>              | <b>23.6%</b>          | <b>~11</b>       |
| 12%               | +0.60                     | 20.5%                 | ~9               |
| 15%               | +0.93                     | 15.9%                 | ~8               |
| 20%               | +1.49                     | 9.4%                  | ~5               |
| 30%               | +2.60                     | 3.0%                  | ~3               |

**Leakage controls.** Because the label looks 63 days forward, a naive contiguous split lets the last 63 training rows “see” the test period through their labels. We therefore impose a *63-day embargo*: the final 63 rows of each training and validation block are purged (López de Prado, 2018). Train, validation, and test are stored as physically separate files and each is windowed *independently*, so no window can straddle a boundary (strictly post-split windowing). The feature scaler is fit on training data only. Figure 3 shows the resulting date ranges; the visible gaps between blocks are the embargo.

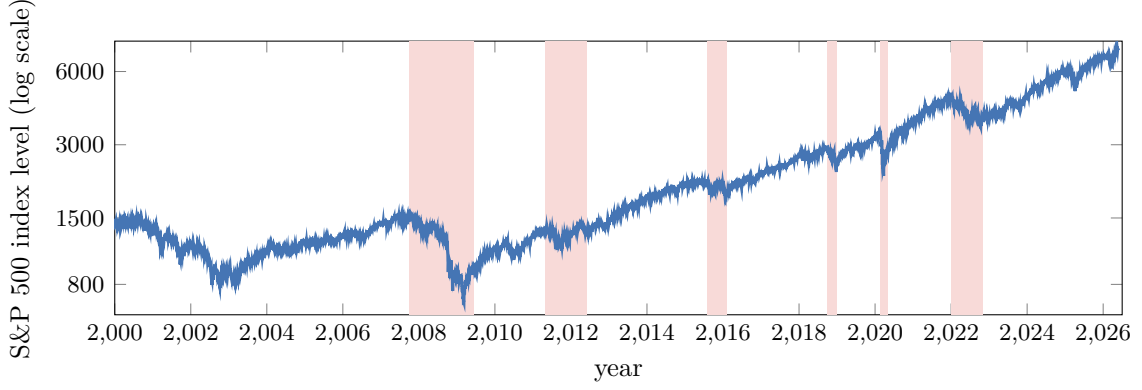
#### 4.3 Model: TFT backbone and the AF recipe

The backbone is the TFT of Lim et al. (2021): per-time-step variable selection networks (VSN) weight the inputs; a sequence-to-sequence LSTM encoder/decoder produces context; static-context GRNs (active only in the panel) inject the market embedding; an interpretable multi-head attention block mixes long-range temporal context; and gated residual networks (GRN) with skip connections carry information around each sub-layer. We replace the quantile-forecast head with a sigmoid classification head for the binary EWS target. Figure 4 shows the backbone (black) with our two modifications (green) and the regime branch we tested and rejected (blue, dashed).

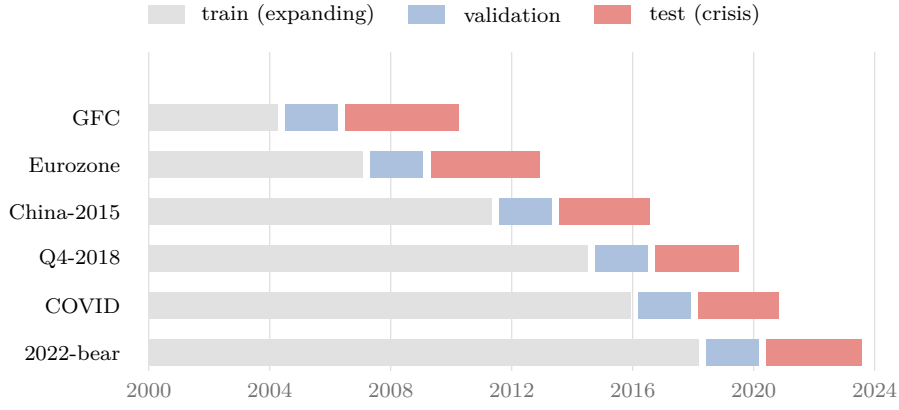
**The AF recipe (our contribution).** Two changes to the *learning problem*, not the backbone topology.

(1) *Auxiliary forward-drawdown head.* A linear head maps each decoder step’s representation  $\mathbf{z}_{t+h} \in \mathbb{R}^s$  to a scalar  $\hat{d}_{m,t,h} = \mathbf{w}^\top \mathbf{z}_{t+h} + b$  predicting the *forward running drawdown* at that step. The target is the realised running drawdown over the decoder horizon,

$$d_{m,t,h} = D_{m,t+h} = \frac{X_{m,t+h}}{P_{m,t+h}} - 1 \leq 0, \quad h = 1, \dots, 63, \quad (2)$$



**Figure 2:** S&P 500 (log scale), 2000–2026, with the six crisis episodes used as test folds shaded: GFC, Eurozone, China-2015, Q4-2018, COVID, 2022 bear.



**Figure 3:** Crisis-anchored walk-forward folds with 63-day embargo. Each row is one fold; grey is the expanding training history, blue the calm validation window, red the crisis test window. The small gaps between blocks are the embargo (purged label-horizon rows). Exact date ranges are listed in Table 13 (Appendix D).

i.e. the same trailing-252-day-peak drawdown used for labelling (Eq. 1), as a raw (unstandardised, signed) fraction. Unlike the sparse binary onset label, this target is defined on *every* forward day (including in-crisis days), supplying a dense regression signal that regularises the shared representation; it is used only in training. The auxiliary loss is the mean squared error over the decoder window,  $\mathcal{L}_{\text{aux}} = \frac{1}{63} \sum_{h=1}^{63} (\hat{d}_{m,t,h} - d_{m,t,h})^2$ .

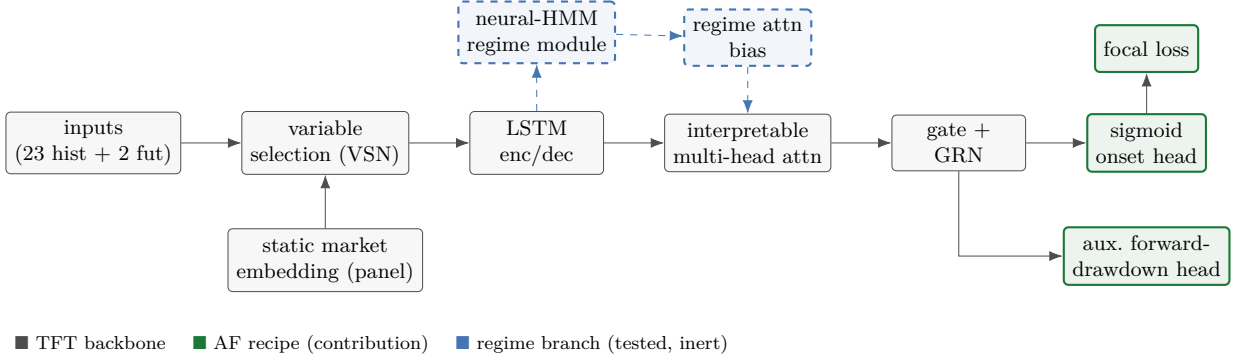
(2) *Focal loss* (Lin et al., 2017) on the onset head, which down-weights easy negatives so the gradient concentrates on the rare positives. With probability  $p = \sigma(\ell)$  for logit  $\ell$ , label  $y$ , and the true-class probability  $p_t = yp + (1-y)(1-p)$ ,

$$\mathcal{L}_{\text{focal}} = -(1 - p_t)^\gamma [w_+ y \log p + (1 - y) \log(1 - p)], \quad (3)$$

i.e. the standard focal loss with the rare positive class up-weighted by  $w_+$ ; the modulating factor  $(1 - p_t)^\gamma$  is clipped below at  $\epsilon = 10^{-6}$  for numerical stability,  $\gamma = 2$  is the focusing parameter, and  $\gamma = 0$  recovers weighted BCE. The full objective is

$$\mathcal{L} = \beta \mathcal{L}_{\text{focal}} + \lambda \mathcal{L}_{\text{aux}} + \gamma_{\text{reg}} \mathcal{L}_{\text{regime}}, \quad (4)$$

with  $\beta = 1$ ,  $\lambda = 0.5$ , and  $w_+$  set near the inverse prevalence (3.4 single-market, 2.5 panel). For the AF model the regime term is off ( $\gamma_{\text{reg}} = 0$ ); the regime variant below sets  $\gamma_{\text{reg}} = 0.3$ . “AFv2” is AF with panel-matched capacity (hidden/state size 128), 60 epochs, learning rate  $5 \times 10^{-4}$ , and per-market normalisation.



**Figure 4:** Model overview. Black: the TFT backbone (Lim et al., 2021). Green: the AF recipe, a multi-task auxiliary forward-drawdown head and focal loss (Eq. 4), which is the contribution. Blue/dashed: the neural-HMM regime module and regime-conditioned attention bias, implemented and ablated but found inert.

**The regime branch (tested, inert).** For completeness we also implement a differentiable neural-HMM *regime module* with  $K = 3$  latent states (calm/stress/crisis) on top of the LSTM hidden states  $\mathbf{h}_{1:T}$ , plus a *regime-conditioned attention bias*. The module produces per-step emission log-likelihoods from a gated residual network and a *time-varying* row-stochastic transition matrix from an MLP,

$$\mathbf{e}_t = \log \text{softmax}(\text{GRN}(\mathbf{h}_t)), \quad A_t = \text{softmax}_{\text{row}}(\text{MLP}(\mathbf{h}_t)) \in \mathbb{R}^{K \times K}, \quad (5)$$

where the MLP bias is initialised to a sticky diagonal (+3 on, −3 off, so each row starts  $\approx [0.9500, 0.0250, 0.0250]$ ). Filtered posteriors  $\boldsymbol{\pi}_t$  are obtained by the differentiable forward algorithm in log-space,

$$\log \alpha_t = \left( \log \sum_{k'} \exp [\log \alpha_{t-1} + \log A_{t-1}] \right) + \mathbf{e}_t, \quad \boldsymbol{\pi}_t = \text{softmax}(\log \alpha_t), \quad (6)$$

initialised at  $\log \alpha_1 = \log \boldsymbol{\pi}_0 + \mathbf{e}_1$ . The regime posteriors over the decoder steps add a learned bias  $b_{\boldsymbol{\pi}_i, \boldsymbol{\pi}_j}$  (a  $K \times K \times H$  tensor over query-regime, key-regime, and head, with  $H$  attention heads) to the attention logits, and a label-smoothed NLL term  $\mathcal{L}_{\text{regime}}$  supervises the posteriors against a three-state regime label (calm = 0; stress = 1 when  $\text{VIX} \geq 25$  and not in crisis; crisis = 2 when the trailing-peak drawdown  $\geq 20\%$ ) and enters Eq. 4. The regime attention is a no-op without the module, so the ablation cells (Section 5.2) are nested rather than fully factorial. Section 5.2 reports that none of this improves on the vanilla backbone.

#### 4.4 Training and evaluation

**The metrics in plain terms.** Because the evidence in Section 5 rests on a few recurring measures, we state them once in words. *Prevalence*  $\pi$  is the fraction of test days that are positive (here  $\pi \approx 0.34$ ); a model that guesses at random scores about  $\pi$ , so  $\pi$  is the “no-skill” floor. *PR-AUC* (area under the precision–recall curve, 0–1, higher better) summarises how well the model ranks true crisis days above calm ones; unlike accuracy it stays meaningful when positives are rare, and it must be read against  $\pi$ , so a PR-AUC of 0.48 against a 0.34 floor is real skill, not near-perfect. *Skill* =  $(\text{PR-AUC} - \pi)/(1 - \pi)$  simply rescales this so 0 = no better than the floor and 1 = perfect. *ROC-AUC* is a related ranking score (chance = 0.50) that, unlike PR-AUC, is insensitive to how rare crises are; we report it for completeness but treat PR-AUC as primary (Saito and Rehmsmeier, 2015). *Calibration* asks whether a stated, say, 30% chance of crisis really does occur about 30% of the time; the *Brier score* (mean squared error of the probabilities, lower better) measures it, and *Platt scaling* is a light post-processing step that rescales raw probabilities to be better calibrated (“Platt” columns are after this step). Scores are *macro-averaged*: computed per crisis fold, then averaged, so each crisis counts equally regardless of its length.

**Protocol.** Optimisation uses AdamW with gradient clipping; the best checkpoint is selected on validation PR-AUC (not validation loss, which rises monotonically here and would select an under-trained epoch). Determinism is hardened (seeded data loaders, exported CUBLAS\_WORKSPACE\_CONFIG, A100-pinned runs). The primary metric is PR-AUC (also reported as skill over the prevalence baseline); we also report ROC-AUC, the Brier score, and post-hoc Platt calibration. Each configuration is run over up to 30 seeds and six folds. *Seeds estimate run-to-run variability of the stochastic neural models; they are not independent observations of market crises.* Primary inference is therefore conducted at the *crisis-episode* level: we average over seeds within each fold and compare models across the six folds (Wilcoxon signed-rank and a block bootstrap that resamples episodes). Seed $\times$ fold tests are reported only as sensitivity diagnostics, never as headline claims, because, with only  $\sim 6$  crisis episodes, statistical power is bounded by episode count, not seed count (Section 5.2).

## 5 Experiments and Results

All numbers below use the leak-free protocol of Section 4.2. We report the single-market study first (backbone, regime ablation, loss/AF study, threshold sensitivity, calibration), then the panel study, then the transfer of the unified recipe.

### 5.1 Backbone: TFT beats LSTM and XGBoost on clean splits

On the embargoed walk-forward splits the TFT backbone is the clear winner among single-market models (Table 2), under the *same* six-fold,  $\approx 30$ -seed protocol as the headline results (so the vanilla TFT figure matches Table 5). An earlier, leakage-contaminated pipeline had reported LSTM ahead of TFT; once the target leak, the all-time-peak labelling bug, and the validation-loss checkpoint bug are fixed, the ranking is TFT > LSTM > XGBoost. This justifies TFT as the backbone.

**Table 2:** Single-market backbone comparison, macro PR-AUC over the six embargoed crisis-anchored walk-forward folds ( $\approx 30$  seeds for the neural models; XGBoost deterministic), prevalence  $\pi = 0.3358$ . Same protocol as Table 5.

| Model              | Macro PR-AUC  |
|--------------------|---------------|
| <b>Vanilla TFT</b> | <b>0.4592</b> |
| LSTM               | 0.3465        |
| XGBoost            | 0.3370        |

**The trees are not crippled.** To ensure the deep advantage is not an artefact of weak baselines, we ran a hyperparameter grid (three configurations each) for XGBoost, Random Forest, and histogram gradient boosting (a LightGBM-style sklearn model). All three trees receive the identical flattened input window as the deep models. Selecting the grid on validation PR-AUC is not possible here: as noted in Section 5.3, four of the six validation splits contain no positive labels, so no validation ranking metric exists for them. We therefore report, for each model and fold, the *best of the three configurations scored on the test fold itself* (Table 3). This is deliberately generous: it is an optimistic upper bound that gives the trees an advantage no honest protocol would grant them, since the configuration is chosen with knowledge of the very set it is scored on. Even so, the best tree (Random Forest, 0.3566) remains far below the deep models (0.4592–0.4869), and the  $\approx 0.13$  gap survives tuning. The trees win only on mild corrections (e.g. Q4-2018) and lose heavily on the severe, volatile crises where the deep models excel (Appendix C). On GFC and Eurozone every tree configuration collapses to the fold prevalence (0.2725 and 0.2843), that is, to a constant prediction with no skill at all.

**Table 3:** Tuned tree baselines vs. deep models, macro PR-AUC over the six folds. Each tree row is the *best of a three-configuration grid scored on the test fold itself*, an optimistic upper bound adopted because four of the six validation splits contain no positive labels and therefore admit no validation ranking metric (Section 5.3); all trees use the same flattened input window as the deep models. The two deep rows are the same conf-M0/AF runs as the headline Table 5 and are *not* test-selected.

| Model                                | Macro PR-AUC  | Macro Brier ↓ |
|--------------------------------------|---------------|---------------|
| <b>AF (deep)</b>                     | <b>0.4869</b> | <b>0.2703</b> |
| Vanilla TFT (deep)                   | 0.4592        | 0.3173        |
| Random Forest (test-selected)        | 0.3566        | 0.3037        |
| XGBoost (test-selected)              | 0.3399        | 0.3866        |
| HistGradientBoosting (test-selected) | 0.3334        | 0.3756        |

## 5.2 Regime awareness is inert

We ablate the regime machinery factorially: M0 (vanilla TFT), M1 (module present but unused), M2 (+ regime aux loss), M3 (+ regime attention), M4 (full RA-TFT). Table 4 (3-fold exploratory run) shows that no single component helps, M1≈M0, M2 and M3 fall *below* M0, and only the full M4 combination edges above vanilla, by a small interaction effect. A 30-seed, six-fold confirmatory run settles the question:

*Full RA-TFT vs. vanilla TFT: macro PR-AUC 0.4630 vs. 0.4592; paired Wilcoxon  $p = 0.4375$  (n.s.).*

**Table 4:** Regime ablation (single-market, exploratory 3-fold, macro PR-AUC raw / Platt). No component helps in isolation; only the full stack (M4) marginally exceeds vanilla, and the effect is not significant under the powered run.

| Cell | Description                   | raw PR-AUC |
|------|-------------------------------|------------|
| M0   | vanilla TFT                   | 0.5274     |
| M1   | regime module present, unused | 0.5221     |
| M2   | + regime auxiliary loss       | 0.5106     |
| M3   | + regime attention bias       | 0.4754     |
| M4   | full RA-TFT                   | 0.5546     |

**Seeds cannot rescue it.** Increasing from 10 to 30 seeds barely moved the  $p$ -value ( $0.0822 \rightarrow 0.0783$  on the three-fold confirmatory run). The reason is structural: the variance that matters is *between crisis episodes* (e.g. COVID is easy, the GFC fold is hard), and with only six episodes the test has little power regardless of seeds. The point is worth stating plainly: *more seeds cannot establish significance for regime awareness; only more independent crises could*, and the data barely stretch to six. That is the central caveat of the whole study. The regime module is a well-motivated idea that does not survive a powered, leak-free test.

## 5.3 What works: the AF recipe

Searching over loss/objective choices on the embargoed splits, focal loss with  $w_+ = 3.4$  is the best loss, and adding the auxiliary forward-drawdown head on top (the AF recipe) is the best configuration overall on every aggregate metric. Table 5 reports the six-fold confirmatory comparison. How we test it matters here: the independent unit of evidence is the crisis episode (six folds), not the random seed, so we report effect sizes and episode-level uncertainty rather than a seed-level  $p$ -value (see *Why not the seed-level  $p$ -value* at the end of this section).

**Table 5:** Single-market headline. *Protocol:* S&P 500, dd10 threshold, six crisis-anchored walk-forward folds,  $\approx 30$  seeds per fold; each cell is the mean over folds of the per-seed mean (macro-averaging), prevalence baseline  $\pi = 0.3358$ . Brier is lower-is-better; skill is  $(\text{PR-AUC} - \pi)/(1 - \pi)$  computed per fold and then macro-averaged. AF = vanilla TFT + auxiliary forward-drawdown head + focal loss. AF lowers Brier in all six folds; the PR-AUC gain is positive but episode-limited (Section 5.3). Post-hoc Platt figures are reported separately below, because Platt could be fitted on only two of the six folds.

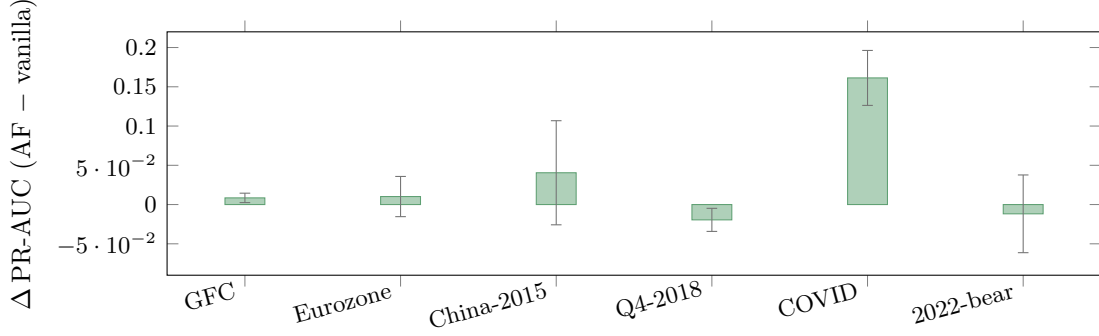
| Model                 | raw PR-AUC    | Brier ↓       | skill         |
|-----------------------|---------------|---------------|---------------|
| M0 (vanilla TFT)      | 0.4592        | 0.3173        | 0.1562        |
| M4 (full RA-TFT)      | 0.4630        | 0.3041        | 0.1655        |
| <b>AF (aux+focal)</b> | <b>0.4869</b> | <b>0.2703</b> | <b>0.1947</b> |

**A note on post-hoc calibration.** Platt scaling is fitted on the validation split, and in four of the six folds (GFC, Eurozone, China-2015 and Q4-2018) that split contains *no positive labels at all*: the validation window falls in a calm stretch, so the positive rate is exactly zero and the scaling cannot be fitted. Platt figures therefore exist only for COVID and 2022-bear, and are reported here for those two folds alone rather than beside the six-fold columns of Table 5, where they would invite a false comparison. Over those two folds the Platt PR-AUC is 0.5451 (M0), 0.6232 (M4) and 0.6560 (AF), and the Platt Brier is 0.2845, 0.2699 and 0.2393 respectively. The apparent lift over the raw six-fold PR-AUC is an artefact of the fold subset, not of the scaling: Platt is monotonic, so it cannot change PR-AUC at all, and indeed on 2022-bear the Platt and raw values agree to six decimal places. The absence of validation positives in four folds is a real limitation of the crisis-anchored design and is revisited in Section 6.1.

**Calibration: a consistent, unambiguous gain.** The clearest result needs no significance test: AF lowers the Brier score in *every one of the six crisis folds* (Table 5, macro 0.2703 vs. 0.3173), with per-fold reductions ranging from 0.0158 (Eurozone) to 0.0906 (China-2015). Dense drawdown supervision yields probabilities a downstream risk rule can trust, and this benefit is uniform across crises rather than concentrated.

**Discrimination: a positive but episode-limited effect.** On PR-AUC, AF improves the macro average by +0.0315 and wins four of six folds, but the effect is *concentrated* in severe, volatile regime-shift crises and is small or slightly negative on mild corrections (Figure 5): COVID +0.1613 and China-2015 +0.0405 dominate, while Q4-2018 (−0.0195) and 2022 (−0.0118) are flat. Throughout, we quote this gain as the *seed-paired* mean difference: we difference AF and vanilla *within* each seed and then average, which cancels the run-to-run noise common to both models and is the quantity the tests below are computed on. It is therefore not identical to subtracting the two macro rows of Table 5 ( $0.4869 - 0.4592 = +0.0277$ ), because seed coverage differs slightly across cells after pre-emption; the two agree to 0.0038. Treating the six crisis episodes as the independent unit, this improvement is *not* statistically significant (Wilcoxon signed-rank over folds  $p = 0.5625$ ; sign test  $p = 0.6875$ ), and a block bootstrap resampling crisis episodes gives a 95% confidence interval for the macro gain of  $[-0.0069, +0.0858]$  that narrowly includes zero (probability of a positive effect 0.9140). We therefore claim a *consistent improvement concentrated where it matters most (real crises)*, not a proven discrimination gain. The paragraph below explains why a seed-level test would overstate this.

**Why not the seed-level  $p$ -value.** A paired Wilcoxon over all  $\approx 170$  seed $\times$ fold pairs returns  $p = 0.0231$ , which looks decisive but is *pseudo-replicated*: thirty random initialisations of the same six crises are not thirty independent crisis observations, and the small  $p$  is driven almost entirely by the single large COVID effect being repeated across correlated seeds. The right unit is the crisis episode, of which we have six; as the regime analysis already noted (Section 5.2), no amount



**Figure 5:** Per-fold improvement of AF over vanilla TFT in PR-AUC (AF – vanilla, differenced within each seed and then averaged over seeds, so that run-to-run noise common to both models cancels). The gain concentrates in the severe COVID (+0.1613) and China-2015 (+0.0405) regime-shift crises and is small or slightly negative on mild corrections (Eurozone +0.0102, GFC +0.0085, 2022 –0.0118, Q4-2018 –0.0195). Error bars are  $\pm 1$  standard error across seeds (run-to-run); single-crisis estimates are highly variable (seed SD up to 0.3511), and the dominant uncertainty is *between* crises; the episode-level bootstrap CI is  $[-0.0069, +0.0858]$  (Section 5.3).

of seeds can manufacture power that only more independent crises could provide. We quote the seed-level number once, here, to flag the discrepancy, and base every discrimination claim on the episode-level statistics above.

**What else we tried and dropped.** Monte-Carlo dropout gave no calibration benefit (slight PR-AUC cost); a regime-conditioned VSN was not significant;  $K = 3$  latent regime states was best ( $K = 5$  significantly worse); and  $\gamma \in \{2, 3\}$  was the best focal focusing range. None of these changed the headline.

#### 5.4 Isolating the contribution: which head does the work?

The AF recipe of Section 5.3 changes two things at once relative to the vanilla baseline: it adds the auxiliary drawdown head, and it swaps weighted BCE for focal loss with  $w_+ = 3.4$ . A comparison against M0 therefore cannot say which ingredient is responsible, and it leaves open a more basic question: if a drawdown regressor is useful, is the classification head needed at all, or would simply *replacing* it suffice? We resolve both with a three-way ablation in which the only thing that varies is which head or heads exist. All three cells share the same backbone with the regime branch explicitly disabled, and every cell that contains a classification term uses the identical focal objective, so the loss is held fixed wherever it is present:

- **CLF-only** — classification head alone, focal loss with  $w_+ = 3.4$ . This is the control the AF-versus-M0 comparison lacks.
- **Both heads** — classification head plus the auxiliary forward-drawdown head; the AF recipe, with the regime branch off.
- **REG-only** — the drawdown head alone. The classification term is removed from the objective entirely ( $\beta = 0$ ), so only the regression loss trains, and the crisis score is read from the regression output.

Scoring REG-only requires care. The regression head emits a drawdown, not a probability, and the sign of its relationship to the crisis label is not fixed: among calm anchors a deeper drawdown indicates elevated risk, whereas over the full test population — which retains windows already inside a crisis — the relationship inverts, because a deep drawdown there marks a crash

that has already happened rather than one about to begin. Assuming either direction would risk manufacturing the result. We therefore fit a one-dimensional logistic map from predicted drawdown to crisis probability on the *training* split alone, so the direction is learned rather than imposed, and we additionally record the raw predictions so that every REG-only model can be re-scored under the opposite sign. Checkpoint selection for this cell uses its own regression objective, since a classification metric would be meaningless for a head that receives no gradient.

**Table 6:** Head ablation. *Protocol:* S&P 500, dd10, six crisis-anchored walk-forward folds, 30 seeds per fold, regime branch disabled throughout, prevalence baseline  $\pi = 0.3358$ . CLF-only and Both-heads share the identical focal objective, so the single difference between them is the auxiliary head. M0 is carried over from Table 5 as the untuned reference. Brier is lower-is-better.

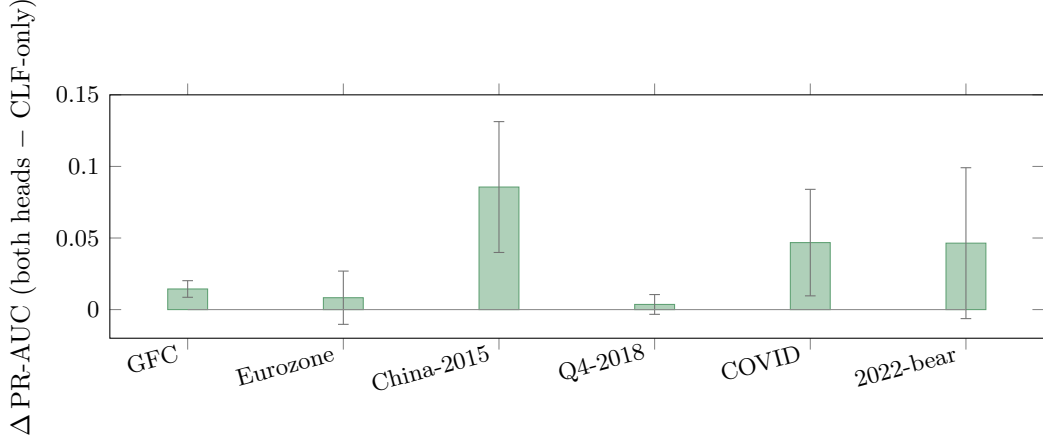
| Configuration                  | raw PR-AUC    | Brier ↓       | skill         |
|--------------------------------|---------------|---------------|---------------|
| M0 (vanilla TFT, weighted BCE) | 0.4592        | 0.3173        | 0.1562        |
| CLF-only (focal, no aux head)  | 0.4521        | 0.2853        | 0.1448        |
| <b>Both heads (AF)</b>         | <b>0.4852</b> | 0.2791        | <b>0.1913</b> |
| REG-only (drawdown head alone) | 0.3649        | <b>0.2494</b> | 0.0365        |

**The auxiliary head, not the loss, carries the recipe.** Holding the objective fixed, adding the drawdown head improves macro PR-AUC by +0.0342 and does so in *all six* crisis folds (Figure 6). Under the episode-level treatment this report argues for throughout, that is a Wilcoxon signed-rank  $p = 0.0312$  — with six episodes, six wins out of six is the smallest two-sided  $p$ -value the test can return — and a block bootstrap over crisis episodes gives a 95% confidence interval for the macro gain of  $[+0.0133, +0.0595]$ , which excludes zero (probability of a positive effect 1.0000). This is a materially stronger statement than the headline of Section 5.3, whose interval  $[-0.0069, +0.0858]$  narrowly included zero, and it is obtained on the conservative unit of analysis rather than the pseudo-replicated seed-level one (the seed $\times$ fold test, quoted once for comparison, gives  $p = 0.0030$ ).

The reason the earlier comparison was weaker is now visible: the focal loss contributes nothing. CLF-only differs from M0 only in the objective, and it is  $-0.0045$  macro PR-AUC *below* it, winning three folds of six (Wilcoxon  $p = 1.0000$ ; seed $\times$ fold  $p = 0.6042$ ). The loss change that the exploratory sweep of Section 5.6 favoured does not survive at six folds and thirty seeds. Attributing the AF gain to “aux + focal” therefore credits an inert ingredient; the auxiliary head accounts for the whole effect. Consistent with Section 5.2, disabling the regime branch inside the AF cell changes macro PR-AUC by 0.0017 (0.4852 against 0.4869), confirming once more that the regime machinery is inert.

**The classification head cannot simply be replaced.** REG-only is far weaker at discrimination: macro PR-AUC 0.3649 against a prevalence baseline of 0.3358, a skill of 0.0365 where the two-head model reaches 0.1913. Re-scoring the identical models under the opposite sign gives 0.4224, still below every configuration that retains a classification head, and the favoured direction is not even consistent across folds — the fitted direction wins on GFC and COVID while the inverted one wins on Eurozone, Q4-2018 and 2022-bear. Even an oracle permitted to choose the better sign per fold in hindsight reaches only  $\approx 0.47$ , short of the two-head model’s 0.4852. The answer to whether a head swap would suffice is therefore no: the drawdown signal is a useful auxiliary target but an unreliable classifier in its own right.

**Why the second head helps.** The per-fold pattern explains the mechanism rather than merely recording it. REG-only outperforms every classification-based configuration on exactly the two folds where those configurations fall *below their own prevalence baseline*: GFC (0.3642 against



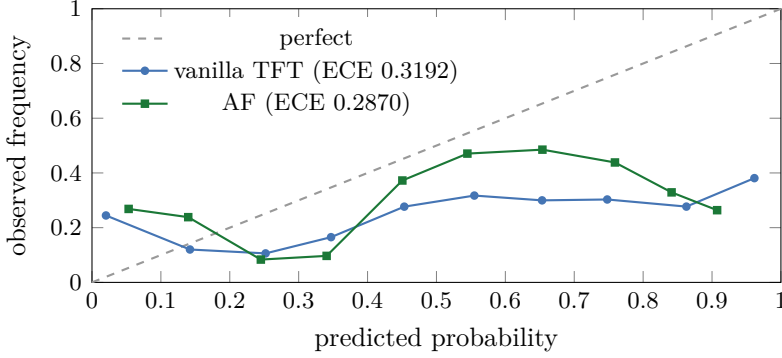
**Figure 6:** Effect of adding the auxiliary drawdown head with the loss held fixed: per-fold PR-AUC difference between the two-head model and the classification-only control, differenced within each seed and then averaged over seeds. Every one of the six crisis folds improves (macro +0.0342; episode-level Wilcoxon  $p = 0.0312$ ; bootstrap CI [+0.0133, +0.0595]). Unlike the AF-versus-vanilla comparison of Figure 5, where the gain was concentrated in COVID and China-2015 and two folds were negative, the benefit here is present everywhere, which is what makes the episode-level test conclusive. Error bars are  $\pm 1$  standard error across seeds; the dominant uncertainty remains *between* crises rather than between runs.

0.2172–0.2335, baseline 0.2725) and Q4-2018 (0.5306 against 0.4019–0.4315, baseline 0.5710). GFC is the extreme case, because its training window contains no positive labels whatsoever (Section 6.1), so the binary target supplies nothing for a classifier to learn from. The drawdown target, by contrast, is defined on every trading day. Where crisis labels are abundant the ordering reverses sharply and the classifier dominates — COVID 0.7935 against 0.5220, China-2015 0.4561 against 0.2828. Dense supervision is thus not a general-purpose substitute for the discriminative signal; it is a complement that supplies gradient precisely in the label-starved regime where sparse crisis targets fail, which is why the two heads together beat either alone. This is the mechanism behind the calibration result of Section 5.3, and it is also why REG-only attains the best Brier score of any cell (0.2494) while being the weakest at ranking: a regression head produces conservative, well-spread probabilities, but conservative probabilities and good discrimination are not the same property.

## 5.5 Calibration and decision usefulness

Because EWS probabilities feed decisions, we examine calibration directly. A second calibration metric corroborates the Brier result of Section 5.3: the per-fold-averaged *expected calibration error* (ECE, ten equal-width bins) is lower for AF than for vanilla TFT (0.2870 vs. 0.3192). Figure 7 shows the corresponding reliability curves: the raw probabilities of *both* models are over-confident (points below the diagonal at high predicted probability), but AF lies closer to the diagonal across the operating range. Calibration is thus the AF recipe’s most consistent benefit: it holds in all six folds (Brier) and on a second, independent metric (ECE).

**Cost-sensitive usefulness.** We also computed the cost-sensitive *usefulness* measure of Sarlin (2013), in which a policymaker weights missed crises against false alarms through a preference parameter  $\mu$  (higher  $\mu$  = more averse to missing a crisis). Writing  $T_1$  for the share of crises missed,  $T_2$  for the share of tranquil periods falsely signalled, and  $P_1 = \pi$ ,  $P_2 = 1 - \pi$  for the unconditional class probabilities, the policymaker’s loss is  $L(\mu) = \mu T_1 P_1 + (1 - \mu) T_2 P_2$ . The benchmark is the loss of the better trivial policy,  $\min(\mu P_1, (1 - \mu) P_2)$ , achieved by always- or never-alarming, and relative usefulness is  $U_r = [\min(\mu P_1, (1 - \mu) P_2) - L(\mu)] / \min(\mu P_1, (1 - \mu) P_2)$ , running from



**Figure 7:** Reliability diagram (per-fold-averaged, raw probabilities). Both models are over-confident relative to the diagonal, motivating the Platt scaling used in all headline results, but AF (green) is closer to perfect calibration than vanilla TFT (blue), consistent with its lower ECE and its lower Brier score in all six folds.

0 (no better than the trivial policy) to 1 (perfect). With the decision threshold chosen at the in-sample optimum, which is an upper bound on achievable usefulness, AF is more useful than the vanilla TFT at every preference setting we examined (Table 7), consistent with its better calibration. Note that under an optimal threshold  $U_r$  is bounded below by zero, because the threshold grid contains the two trivial policies. The measure itself is not uncontested: Alessi and Detken (2014) argue that normalising by the unconditional crisis probability leaves  $U_r$  sensitive to small perturbations of  $\mu$  and awkward to use for policy. That criticism is why Table 7 reports three preference settings rather than one. The *level* of  $U_r$  should not be read too literally, but the ordering is stable across all three settings, and it is the ordering, not the level, that the comparison rests on.

**Table 7:** Sarlin relative usefulness  $U_r$  (per-fold-averaged over the six folds, ten seeds per fold, in-sample optimal threshold, an upper bound). Higher is better;  $\mu$  is the policymaker’s aversion to missed crises.

| Model            | $\mu = 0.5$   | $\mu = 0.7$   | $\mu = 0.9$   |
|------------------|---------------|---------------|---------------|
| M0 (vanilla TFT) | 0.1365        | 0.2674        | 0.2562        |
| <b>AF</b>        | <b>0.2008</b> | <b>0.3310</b> | <b>0.3224</b> |

More importantly, the *out-of-sample* version, choosing the threshold on the calm validation window and applying it to the crisis test window, is *not robustly estimable here*: the pre-crisis validation windows contain too few onsets to fix a transferable threshold, and where one exists it transfers poorly across the calm-to-crisis prevalence shift. That is a finding in its own right about single-market EWS: a fixed alarm threshold set in quiet times does not carry into a crisis, which is exactly why *calibrated probabilities* (where AF helps) beat a single hard threshold.

## 5.6 Threshold sensitivity

The single-market advantage is specific to dd10. Repeating the comparison at  $\tau \in \{12, 15, 20\}\%$  shows the field collapsing toward a tie as the threshold rises and episodes become too rare to learn from or evaluate on (Table 8). This is expected (at dd20 a fold contains  $\sim 1$  onset) and it motivates the dd10 operating point. We report it openly as a limitation rather than tuning the threshold to the result.

*This is a separate, earlier sweep* (a three-fold run of the regime-aware model on a different harness), used only to study how the threshold choice affects the deep-vs-tree gap; its “best deep” column is *not* the six-fold AF headline of Table 5 (whose macro PR-AUC is 0.4869) and the two should not be compared directly. The point of the table is the *trend* across thresholds, not the

level.

**Table 8:** Threshold-sensitivity sweep (macro PR-AUC, three-fold, regime-aware deep model; a diagnostic run distinct from the six-fold AF headline of Table 5). “Best deep” is the strongest deep configuration in this sweep at each threshold. The deep-model advantage is strongest at dd10 and erodes as the threshold rises and events become scarce. (XGBoost not available at dd12/dd20 in this sweep.)

| Threshold | best deep (this sweep) | LSTM        | XGBoost |
|-----------|------------------------|-------------|---------|
| dd10      | <b>0.56</b>            | 0.43        | 0.33    |
| dd12      | 0.35                   | <b>0.39</b> | –       |
| dd15      | <b>0.36</b>            | 0.33        | 0.35    |
| dd20      | 0.34                   | <b>0.36</b> | –       |

### 5.7 Panel: the tuned recipe is competitive with XGBoost

**Panel protocol.** The panel pools the nine indices of Section 3 into a single multi-entity dataset, with market identity as a static categorical consumed by TFT’s static encoder (a learned market embedding). Unlike the single-market study, the panel uses a *single chronological split shared across all markets*, train to 2015-12-31, validation to 2018-12-31, test 2019–2026, rather than crisis-anchored walk-forward folds; a 63-day embargo is applied per market at each boundary, the feature scaler is fit on the training span only, and windows are built within each split. Metrics are pooled over all (market, day) test observations. We first report this pooled chronological split because it directly tests whether more entities lift deep models above trees at all; we then subject the same comparison to a five-fold panel walk-forward robustness check (Table 10).

**A caveat on “ $\approx 110$  episodes”.** The panel contains more crisis *onsets* than the single market, but they are *not* independent: the nine indices are highly correlated and most crash together (e.g. COVID is a crisis in all nine within the same weeks), so the effective number of independent episodes in the 2019–2026 test span is far smaller than the raw onset count. The panel therefore buys representational variety and training signal more than it buys independent statistical power, and we treat its significance tests with the same caution as the single-market ones.

The nine-market panel is harder in absolute terms, pooling heterogeneous markets lowers PR-AUC relative to the single-market S&P 500, but it activates the static pathway and gives the deep models more to fit. A first pass at single-market-sized capacity left them *under-fit* and XGBoost led; matching capacity and training budget to the larger data (the AFv2 configuration) closes that gap and brings the tuned deep model level with or marginally ahead of XGBoost (Table 9).

**Table 9:** Panel results, pooled test set (2019–2026), prevalence  $\pi = 0.2996$ . AFv2 ( $n = 32$  seeds) = AF + capacity 128 + 60 epochs + lr  $5 \times 10^{-4}$  + per-market normalisation; deep baselines  $n = 12$ –14, XGBoost deterministic. Column-best in bold: AFv2 leads on PR-AUC and Brier; the per-market-normalised vanilla model edges ROC-AUC. Differences among the top group are small (see text).

| Model                      | raw PR-AUC    | ROC-AUC       | Brier ↓       |
|----------------------------|---------------|---------------|---------------|
| <b>AFv2 (tuned recipe)</b> | <b>0.3846</b> | 0.5940        | <b>0.2705</b> |
| M0 + per-market norm       | 0.3767        | <b>0.5972</b> | 0.3470        |
| XGBoost                    | 0.3728        | 0.5854        | 0.2730        |
| AF (global norm)           | 0.3650        | 0.5730        | 0.2804        |
| Vanilla TFT (M0)           | 0.3433        | 0.5519        | 0.3679        |
| Full RA-TFT (M4)           | 0.3373        | 0.5520        | 0.3661        |
| LSTM                       | 0.2966        | 0.4896        | 0.4775        |

At 32 seeds, AFv2 has the best PR-AUC (0.3846) and Brier (0.2705) of any model and a competitive ROC-AUC, but the margins within the top group are small: AFv2 vs. the per-market-normalised vanilla baseline is +0.0079 PR-AUC (paired Wilcoxon  $p = 0.1514$ ), and AFv2 vs. XGBoost is +0.0118. AF still clearly beats LSTM ( $p = 0.0010$ ), and the full RA-TFT remains indistinguishable from vanilla ( $p = 0.95$ ): the regime module is inert again, now with a market embedding and many more onsets, a further confirmation of the single-market finding. The attribution is that what lifts the deep models is *capacity, training budget, and the retuned loss* together with per-market normalisation (which helps the vanilla baseline most). The measured conclusion is that a properly tuned TFT-based deep model is competitive with, and on PR-AUC and calibration marginally ahead of, gradient boosting on this panel. That is neither the large deep-beats-trees reversal an earlier random-holdout analysis had suggested, nor the trees-dominate verdict of the single-market literature. Bootstrapping the 32 AFv2 seeds gives a 95% confidence interval for its mean PR-AUC of  $[0.3639, 0.4048]$ , which spans XGBoost’s 0.3728; the probability that AFv2’s mean exceeds XGBoost is 0.8686.

**Walk-forward robustness.** To check that this is not an artefact of the single 2019–2026 split, we repeated the panel comparison on five crisis-anchored walk-forward folds (expanding train, calm validation, crisis test, per-market embargo), pooled across the nine markets (Table 10). The conclusion is unchanged: AFv2 has the best macro PR-AUC (0.3476) and Brier, marginally ahead of XGBoost (0.3331) and clearly ahead of the per-market-normalised vanilla baseline (0.2878), winning three of the five folds; XGBoost edges ROC-AUC. The walk-forward thus corroborates the single-split result: the tuned deep recipe is competitive with, and slightly ahead of, gradient boosting on the panel, while the lower absolute PR-AUC confirms that the crisis-anchored folds are harder than the pooled split.

**Table 10:** Panel *walk-forward* robustness: macro over five crisis-anchored folds (expanding train, per-market 63-day embargo), pooled across nine markets. AFv2/M0pm averaged over seeds; XGBoost deterministic.

| Model                      | macro PR-AUC  | ROC-AUC       | Brier ↓       |
|----------------------------|---------------|---------------|---------------|
| <b>AFv2 (tuned recipe)</b> | <b>0.3476</b> | 0.5182        | <b>0.2737</b> |
| XGBoost                    | 0.3331        | <b>0.5444</b> | 0.3002        |
| M0 + per-market norm       | 0.2878        | 0.4265        | 0.4191        |

## 5.8 One unified recipe

Finally, the panel-winning AFv2 configuration transfers *back* to the single-market task without harm. In a dedicated transfer experiment (15–30 seeds), AFv2 reaches macro PR-AUC 0.4953 versus 0.4869 for the lean AF variant and 0.4592 for vanilla, and is statistically tied with lean AF ( $p = 0.3282$ ). (The vanilla and lean-AF figures coincide with the headline six-fold values in Table 5.) AFv2 wins the high-volatility folds (COVID +0.04, 2022 +0.09) and dips slightly on the smallest folds, a mild capacity-versus-data trade-off that nets to a wash. The practical conclusion is a *single* recommended model: AFv2 wins the panel and ties the best single-market model, while a leaner AF variant (hidden 64, 40 epochs) is an equivalent, cheaper option when only a small single-market dataset is available.

## 6 Discussion

**Why the regime module fails and the aux head succeeds.** Both add parameters that “know about crises”, but they differ in where the supervisory signal comes from. The regime module must *infer* latent states from the same scarce labels that already bottleneck the task; it

adds capacity without adding information. The auxiliary head, by contrast, supplies a *new, dense* signal, the forward drawdown magnitude is defined on every day, which regularises the shared representation and is exactly what a sparse-label problem needs. Focal loss then re-allocates gradient toward the rare positives. The lesson generalises: in low-event regimes, adding *information* (denser supervision) beats adding *capacity* (architectural machinery).

**Calibration is the most reliable benefit.** Across both studies the AF recipe’s clearest, most consistent edge is on the Brier score. For an EWS whose outputs feed risk decisions, well-calibrated probabilities are arguably more valuable than a marginal ranking gain, and this benefit is robust where the PR-AUC gain is concentrated.

**Evaluation manufactures conclusions.** The preliminary version of this project reported, under a random holdout, that a regime-aware TFT beat XGBoost on the panel. Under leak-free temporal CV that result does not reproduce. The reversal is itself a finding: in financial ML the protocol (embargo, post-split windowing, prevalence-relative metrics, paired tests) is not bookkeeping; it decides what the experiment measures.

## 6.1 Limitations

- **Few episodes.** Even pooled, the number of *independent* crisis episodes is small and the episodes are cross-correlated (markets crash together), bounding statistical power. We report effect sizes and *p*-values together and refrain from over-claiming.
- **Overlapping windows.** Stride-1 windows create autocorrelated samples; bootstrap confidence intervals on PR-AUC should be read as indicative. Block-bootstrap CIs and a non-overlapping (stride-63) evaluation are natural robustness checks (future work).
- **Single-path walk-forward.** A combinatorial purged cross-validation (CPCV) would give a distribution of outcomes rather than one path; we defend the single path on the grounds that, with  $\sim 6$  highly correlated episodes, CPCV paths are non-independent and add little (López de Prado, 2018).
- **Threshold and domain.** Conclusions are specific to dd10 and to daily equity-index drawdown; the supporting EWS literature is largely cross-country macro-banking, so it validates our *principles* more than our exact setup.

## 7 Legal, Social, Ethical and Professional Issues

This project builds a predictive model of financial distress from licensed market data. It therefore raises questions about data rights and intellectual property, about the harm an over-trusted warning system could do, and about the obligation to report what was found rather than what was hoped for. This chapter sets out how those questions were handled, with reference to the *British Computer Society* (BCS) Code of Conduct (BCS, 2022) and the *Institution of Engineering and Technology* (IET) Rules of Conduct (IET, 2023).

### 7.1 Legal and intellectual property

The series underlying every result are daily Bloomberg data, licensed to the institution and not free to redistribute. Two consequences follow. First, *no Bloomberg data appears in the public project repository*: the raw pull and every real processed dataset derived from it are excluded. So that the pipeline can nonetheless be executed and inspected, the repository ships clearly labelled

*mock* datasets in their place — surrogates generated from the licensed data by heavy seeded stochastic perturbation, whose values correspond to no real market quote, cannot recover the original series, and are documented in the repository as unusable for any purpose other than running the code. Second, the data pipeline is documented (Section 3) in enough detail that the real datasets can be rebuilt from an equivalent licensed source; together with the mock data, that is the form of reproducibility available when the inputs themselves cannot be shared. All results in this report were produced on the real data. The macro-financial series taken from the Federal Reserve Economic Data service carry no such restriction.

The modelling code is our own and is released under the MIT licence; that grant covers the source code only and does not extend to the Bloomberg-derived data. Third-party libraries (PyTorch, scikit-learn, XGBoost, NumPy, pandas and SciPy) are used under their permissive licences, are not redistributed, and are declared with pinned versions. This project also builds on published work: the TFT backbone (Lim et al., 2021), the focal loss (Lin et al., 2017), the purging-and-embargo protocol (López de Prado, 2018), and the regime-switching literature (Hamilton, 1989; Ang and Bekaert, 2002). Each is implemented from the published description and cited at every point of use, in line with the BCS requirement to have due regard for the legitimate rights of third parties.

## 7.2 Social impact and risk

An early-warning system is a decision aid, and its errors are asymmetric in cost. A false alarm may prompt an institution to de-risk unnecessarily, at a price ultimately borne by savers and pension holders; a missed onset leaves exposure in place precisely when it matters most. Neither error is abstract, and both argue against presenting a research model as though it were an operational one.

We therefore do not claim this model is deployment-ready, and the evidence in this report supports that restraint rather than sitting awkwardly beside it. The discrimination gain is not significant at the episode level (Section 5.3); the out-of-sample decision threshold is not robustly estimable from calm validation windows (Section 5.5); and the raw probabilities of both models are over-confident before calibration. Stating those limits plainly is the substance of the BCS duty not to misrepresent the performance of a system, rather than a caveat appended to it.

A subtler risk is reflexivity. A crisis-warning model is not a neutral observer of the system it forecasts: were such a signal widely adopted and acted upon, the resulting de-risking could itself help propagate the downturn being predicted. This is a reason to prefer calibrated probabilities with disclosed uncertainty over a binary alarm, and it strengthens the case (Section 5.5) for treating calibration rather than ranking as the property that matters in deployment. The project uses no personal data and involves no human subjects; the inputs are aggregate market prices, so the privacy dimension of the BCS public-interest principle is not engaged here.

## 7.3 Professional conduct

The BCS Code of Conduct requires members to accept professional responsibility for their work and not to misrepresent or withhold information on its performance; the IET Rules of Conduct place comparable duties of integrity and public interest on engineers. Three decisions in this project were taken on those grounds, and each cost something.

First, the preliminary version of this work reported that a regime-aware TFT beat gradient boosting under a random holdout. That result does not survive leak-free temporal validation. It is reported here as falsified (Section 6) rather than quietly dropped, because the reversal is evidence about the protocol, and withholding it would misrepresent how the final conclusion was reached.

Second, a paired test over seed-fold pairs returns  $p = 0.0231$  for the headline comparison,

which could be presented as significant. It is pseudo-replicated: thirty initialisations of the same six crises are not thirty independent observations of crises. It is therefore reported only as a diagnostic (Section 5.3), and the honest episode-level test, which does not reach significance, is reported as the result. Choosing the weaker but correct number over the stronger but misleading one is the concrete form the integrity requirement takes here.

Third, the central hypothesis of the project failed. Reporting the regime module as inert, together with the ablation that establishes it, is of more use to a reader than a narrative in which the initial idea happened to succeed.

All sources are cited, all reused ideas attributed, and the statistical claims are reproducible from the archived code and configurations (Appendix A). Claims of competence are bounded by the limitations set out in Section 6.1.

## 8 Conclusion

We set out to test whether regime awareness, a crisis-specific inductive bias bolted onto a Temporal Fusion Transformer, helps predict equity crises. It does not. Under a leak-free protocol the regime-aware model never separates from a plain TFT, on the single market and on the panel alike. The change that does earn its place is a modest one: a multi-task forward-drawdown head and a focal loss. That recipe calibrates better in every crisis fold, lifts ranking accuracy where the crises are severe, and, scaled up, edges gradient boosting on the nine-market panel while transferring back to the single market unchanged. The lasting lesson is about evaluation as much as modelling. The protocol that exposes the regime module as inert is also what makes the recipe’s modest but real gains believable; and it shows that when crises are scarce, feeding the model more signal beats giving it more parameters.

Measured against the objectives of Section 1.3: the leak-free protocol was built and, in falsifying our own earlier result, proved its worth (objective 1); the backbone comparison was run against tuned trees (objective 2); the regime branch was implemented in full and ablated component by component (objective 3); dense auxiliary supervision and a focal loss were introduced and became the contribution (objective 4); the recipe was carried to a nine-market panel (objective 5); and calibration was evaluated alongside discrimination, where it proved the more robust gain (objective 6).

### 8.1 Lessons learned

Four lessons stand out, and the first three were learned the hard way.

**The protocol is the experiment.** The largest swing in results across this project came from no model change at all. Fixing a target leak, replacing an all-time-high drawdown definition with a trailing-peak one, and selecting checkpoints on the evaluation metric rather than on validation loss moved the numbers further, and in a different direction, than any architectural idea tried afterwards. A result obtained before the protocol was sound was not a weaker result; it was a different question’s answer.

**Statistical power is a property of the data, not of the compute budget.** Considerable time was spent raising seed counts from ten to thirty in the hope of resolving a marginal  $p$ -value. It moved from 0.0822 to 0.0783, because the independent unit of evidence is the crisis episode and there were still only six of them. Recognising which quantity actually binds an experiment, before spending a compute allocation on the one that does not, is a lesson that generalises well beyond this project.

**Underfitting can impersonate a scientific conclusion.** The first panel run appeared to show gradient boosting beating the deep models, a tidy result consistent with the literature and

therefore easy to believe. It was an artefact of running a single-market-sized model on a dataset four times larger. A finding that flatters the prior literature deserves the same scepticism as one that contradicts it.

**Calibration deserves parity with discrimination.** We began, by habit, optimising and reporting a ranking metric. The result that survived scrutiny best was the calibration improvement: uniform across all six folds, and the one an actual risk desk could use. It was almost an afterthought in the original evaluation design.

## 8.2 Future work

Several robustness checks suggested by the rare-event-EWS literature have been incorporated into this work: per-window-prediction reliability diagrams and the Sarlin (2013) usefulness measure (Section 5.5), a tuned multi-model tree-baseline comparison (Table 3), block-bootstrap confidence intervals, and a panel walk-forward in addition to the pooled split (Table 10). The remaining directions are:

1. **More crisis episodes.** Extending the panel (more indices, sectors, or asset classes, each a TFT entity) is the highest-value way to gain statistical power and to give the static embedding more to learn.
2. **Combinatorial purged CV** to convert the walk-forward paths into a full distribution of outcomes and apply a trial-count / probability-of-backtest-overfitting correction; with  $\sim 6$  highly correlated episodes its added value is limited, so we defended the single-path design here.
3. **Gradient-boosting breadth.** The tree field used XGBoost, Random Forest, and a histogram GBM; bespoke LightGBM/CatBoost pipelines with tree-appropriate feature engineering (rather than the flattened window the deep models use) would be a yet stronger baseline.
4. **Auxiliary-task design.** The aux head is the lever that works; multi-horizon drawdown targets, a volatility auxiliary, or a quantile drawdown head are natural extensions of the dense-supervision idea.
5. **Backbone variants.** A Mamba/state-space or TCN encoder in place of the LSTM, ablated under the same protocol, would test whether the backbone choice matters once the AF recipe is fixed.
6. **Regime, revisited.** The regime module did not help here; whether an *externally supervised* regime label (e.g. NBER/credit-cycle states) rather than a self-inferred one would help is an open question.

## A Reproducibility and Compute

All experiments were run on the King’s College London CREATE high-performance cluster (NVIDIA A100 GPUs) via Slurm array jobs. Determinism was hardened by seeding Python, NumPy, and PyTorch (including the data-loader generator), exporting `CUBLAS_WORKSPACE_CONFIG` before CUDA initialisation, pinning runs to a single GPU model, and enabling deterministic algorithms where available; residual non-determinism (e.g. cuDNN LSTM kernels) is absorbed by multi-seed averaging. The best checkpoint per run is selected on validation PR-AUC (not validation loss, which rises monotonically here and would select an under-trained epoch). Post-hoc Platt scaling is fit on validation predictions only. Crisis labels, features, and walk-forward folds are produced by the project’s data pipeline from raw Bloomberg series; the modelling code (TFT

backbone, AF head, regime module, loss functions), experiment scripts, and the figure-generation script (`scripts/make_figures.py`, which writes the `.dat` files behind every plot) are in the project repository at <https://github.com/sarpvulas/crisis-ews-tft>. The data are licensed Bloomberg series and cannot be redistributed; the repository instead includes clearly labelled mock surrogate datasets so the pipeline runs end-to-end without Bloomberg access (Section 7), and the full processing code and configuration are provided so the real datasets can be rebuilt from an equivalent licensed source. All results in this report were produced on the real data.

## B Hyperparameters

Table 11 lists the settings. The “lean” single-market configuration is the cheaper variant referenced in Section 5.8; AFv2 is the capacity-matched configuration used on the panel and as the unified recipe.

**Table 11:** Model and training hyperparameters.

| Setting               | lean (single-market) | AFv2 (panel/unified) |
|-----------------------|----------------------|----------------------|
| Encoder length        | 252                  | 252                  |
| Decoder length        | 63                   | 63                   |
| Hidden / state size   | 64                   | 128                  |
| Attention heads       | 4                    | 4                    |
| Dropout               | 0.1                  | 0.1                  |
| Optimiser             | AdamW                | AdamW                |
| Learning rate         | $10^{-3}$            | $5 \times 10^{-4}$   |
| Epochs                | 40                   | 60                   |
| Focal $\gamma$        | 2                    | 2                    |
| Positive weight $w_+$ | 3.4                  | 2.5                  |
| Aux. weight $\lambda$ | 0.5                  | 0.5                  |
| Regime states $K$     | 3                    | 3                    |
| Per-market norm       | —                    | yes                  |
| Calibration           | Platt                | Platt                |

## C Full per-fold results

Table 12 reports per-fold raw PR-AUC (mean  $\pm$  standard deviation across seeds) for the three single-market models, and the raw Brier score for the vanilla and AF models. The large per-fold seed standard deviations (e.g. China-2015, 2022) show that single-crisis estimates are intrinsically noisy; the constraint this places on statistical power is discussed in Section 5.2. AF lowers the Brier score in every fold.

## D Fold date ranges

Table 13 gives the exact train/validation/test date ranges of the six single-market walk-forward folds (Figure 3). All folds share the training start 2000-01-03 (expanding window); the gaps between blocks are the 63-day embargo. The final column is the number of positive (onset) labels in the test window.

**Table 12:** Per-fold single-market results ( $\approx 30$  seeds/fold, dd10, 63-day embargo). Raw PR-AUC as mean $\pm$ SD across seeds; raw Brier (lower better) for M0 and AF.

| Fold       | raw PR-AUC (mean $\pm$ SD) |                    |                           | Brier $\downarrow$ |               |
|------------|----------------------------|--------------------|---------------------------|--------------------|---------------|
|            | M0                         | M4                 | AF                        | M0                 | AF            |
| GFC        | 0.2172 $\pm$ .0412         | 0.2311 $\pm$ .0620 | 0.2257 $\pm$ .0468        | 0.2537             | <b>0.2254</b> |
| Eurozone   | 0.3135 $\pm$ .1377         | 0.3280 $\pm$ .1263 | 0.3238 $\pm$ .1171        | 0.2827             | <b>0.2669</b> |
| China-2015 | 0.5409 $\pm$ .1998         | 0.4957 $\pm$ .2217 | 0.5748 $\pm$ .2295        | 0.2781             | <b>0.1875</b> |
| Q4-2018    | 0.4315 $\pm$ .0594         | 0.4478 $\pm$ .0779 | 0.4120 $\pm$ .0574        | 0.5808             | <b>0.5352</b> |
| COVID      | 0.7353 $\pm$ .1681         | 0.7652 $\pm$ .1617 | <b>0.8966</b> $\pm$ .0981 | 0.2297             | <b>0.1941</b> |
| 2022 bear  | 0.5169 $\pm$ .2028         | 0.5103 $\pm$ .1329 | 0.4883 $\pm$ .1958        | 0.2787             | <b>0.2127</b> |
| Macro      | 0.4592                     | 0.4630             | <b>0.4869</b>             | 0.3173             | <b>0.2703</b> |

**Table 13:** Single-market walk-forward fold date ranges (training start 2000-01-03 throughout). “Test +” is the count of onset-positive days in the test window.

| Fold       | Train end  | Val start  | Val end    | Test start | Test end   | Test + |
|------------|------------|------------|------------|------------|------------|--------|
| GFC        | 2004-04-02 | 2004-07-01 | 2006-04-04 | 2006-07-03 | 2010-04-01 | 92     |
| Eurozone   | 2007-01-31 | 2007-05-01 | 2009-02-02 | 2009-05-01 | 2012-11-30 | 126    |
| China-2015 | 2011-05-03 | 2011-08-01 | 2013-05-03 | 2013-08-01 | 2016-08-01 | 63     |
| Q4-2018    | 2014-07-03 | 2014-10-01 | 2016-07-05 | 2016-10-03 | 2019-06-28 | 63     |
| COVID      | 2015-12-02 | 2016-03-01 | 2017-12-01 | 2018-03-01 | 2020-10-30 | 63     |
| 2022 bear  | 2018-03-05 | 2018-06-01 | 2020-03-03 | 2020-06-01 | 2023-08-01 | 63     |

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